

FRIDA CARDONA

MACHINE LEARNING ENGINEER | COMPUTER VISION | DEEP LEARNING
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PROFESSIONAL SUMMARY

Machine learning engineer specialized in computer vision and anomaly detection systems deployed in real-world industrial environments. Experienced in developing end-to-end ML pipelines including data collection, annotation, model training, optimization and production deployment. Strong hands-on experience with PyTorch, YOLO, and production ML systems integrated with automation and embedded hardware.

KEY SKILLS

Machine Learning & Deep Learning | Computer Vision Systems | Retrieval-Augmented Generation (RAG) | LLM Integration & Embeddings
| Machine Learning Deployment | Industrial AI Applications

TECHNICAL SKILLS

- **Machine Learning & Deep learning:** PyTorch, Computer vision, Object detection (YOLOv8), Anomaly detection (PaDiM, Anomalib), Regression models, CNNs.
- **LLMs & AI Systems:** Retrieval-Augmented Generation (RAG), Embeddings, Prompt Engineering, Natural Language Processing (NLP), FAISS, Ollama, OpenAI API, NLP.
- **ML Deployment & Optimization:** Python, OpenVINO, CUDA, Real-time inference systems, Azure Container Apps, Azure Static Web Apps
- **Data Preprocessing:** Pandas, NumPy, Data preprocessing, Dataset creation and annotation.
- **Software Development:** Python, Git, REST APIs, MicroPython, C#.

WORK EXPERIENCE

Personal Project: AI Me – AI Portfolio Assistant (RAG-based)

Developed an AI-powered portfolio assistant that answers questions about my professional experience using Retrieval-Augmented Generation (RAG). URL: <https://brave-water-02b185810.1.azurestaticapps.net/>

- Designed and implemented a full RAG pipeline using FAISS and SentenceTransformers for semantic search over personal experience transcripts
- Built a FastAPI backend containerized with Docker and deployed on Azure Container Apps
- Developed a React + Vite frontend deployed on Azure Static Web Apps
- Integrated vector embeddings generated from structured transcripts of professional experience
- Optimized inference latency by preloading models and embeddings during container startup
- Implemented full-stack communication between frontend and backend APIs
- Managed environment variables and deployment workflows across cloud services

Senior Automation Software Engineer | PEGATRON MEXICO | Ciudad Juarez, Mexico

Feb 2025 – Current

Develop machine learning and computer vision systems deployed in production environments.

- **GenAI Maintenance Assistant – Industrial RAG System**
GitHub: <https://github.com/fridacardona-frida/genai-maintenance-assistant>
Developed Retrieval-Augmented Generation (RAG) system for industrial troubleshooting assistance, implemented semantic search using SentenceTransformers embeddings and FAISS vector database, build backend API using FastAPI and Uvicorn with /ask endpoint for contextual query processing, integrated local LLM inference using Ollama for privacy-preserving deployment, designed frontend chat interface using Vite and TypeScript,
- **Anomaly Detection System**
Developed deep learning anomaly detection system using Anomalib PaDiM model with backbone ResNet50. Built complete training and inference pipeline, trained model on real industrial dataset, optimized inference using OpenVINO and CUDA, Integrated system into production line for real-time defect detection, Prevented defective units from reaching customers.
- **Computer Vision Inspection System**
Developed and deployed YOLOv8 object detection model for industrial inspection of a manual process following the operator steps with MediaPipe, Collected and annotated custom production dataset, Trained and optimized model using PyTorch, Integrated model into Python desktop application developed with CTK for real time inference, deployed system into production environment, reduced scrap rate by 10%
- **Embedded Machine Learning System**
Developed regression model for real-time sensor calibration, Collected and processed sensor dataset, trained regression model to convert sensor data into physical measurements, deployed model into embedded ESP32 system, integrated model with backend API.

Automation Engineer | PEGATRON MEXICO | Ciudad Juarez, MexicoApr 2024 – Feb 2025

Developed automation, robotics and production systems.

- Automation and Predictive Systems
 - Developed predictive glue waste models (Sci-kit learn).
 - Glue consumption analysis.
 - Defective production diagnostics related to machines failures.
- Industrial Robot Programming and Vision Integration
 - Programmed and deployed industrial robots (EPSON SCARA robots, EPSON 6-axis robots) using EPSON RC+ software.
 - Programmed pick and place routines.
 - Integrated robot vision system for position correction, configured vision-guided alignment.

CNC Programmer | HOLMAR | Ciudad Juarez, ChihuahuaSep 2023 – Apr 2024

- Mechanical design and manufacturing
 - Designed mechanical parts using SolidWorks, Generated CNC G&M programs, Manufactured precision components

Jr Electronics Engineer (Intern) | BIOBOT Farm | Chihuahua, ChihuahuaAug 2021 – Aug 2022

IoT agricultural automation systems

Worked with: Arduino, LoRa, GPS, GPRS, Bluetooth

Responsibilities: Hardware integration, Sensor installation, Software debugging, Field troubleshooting

EDUCATION CERTIFICATIONS & AWARDS

Microsoft AI & ML Engineering Professional Certificate Coursera	2025-Current
BACHELOR'S DEGREE Mechatronics Engineer GPA 4.0 TECNM Ciudad Jimenez Best GPA of the course	2018 – 2022
TECHNICAL DEGREE Programming Technician CBTIS 138 Best GPA of the course	2015 – 2018
EXPOCIENCES CHIHUAHUA 1 ST PLACE	2019, 2021